

★ WEEK 15 · PART 2 of 3

Sampling States and Parameters Together

FFBS, the Bayesian Hedge Ratio, and Stochastic Volatility

Two-block Gibbs · forward-filter backward-sample · TVP-SV

Now the machinery. A two-block Gibbs sweep alternates between states and parameters; FFBS draws a whole state path at once. We put it to work on a time-varying hedge ratio, then add a breathing volatility.

2

PART 2 · SAMPLING THE MODEL

From Joint Posterior to Draws

The two-block Gibbs sweep, forward-filter backward-sample for whole state paths, the Bayesian time-varying hedge ratio with credible bands, and TVP-SV when the noise level itself moves.

From Model to Machinery

Part 1 defined it; Part 2 samples it

1

We want a joint posterior

Part 1: infer the hidden state path and the parameters together, and get an honest fan instead of a confident line.

2

But sampling it looks daunting

The state path can have hundreds of entries, all correlated, entangled with the parameters. How do we draw from that?

3

Split it into two easy blocks

Draw the states given the parameters, then the parameters given the states. Each half is easy; alternating explores the joint posterior.

Part 2 in one line: alternate states and parameters, draw whole paths with FFBS, apply it to the pair.

Sampling States and Parameters Together

A two-block Gibbs sweep for dynamic models

The joint posterior over a long state path and the parameters is a tangle — but it untangles if you condition. Given the parameters, the states are a clean linear-Gaussian problem; given the states, the parameters are a simple regression. Gibbs alternates between the

two

The hard version

- States and parameters are correlated and coupled.
- Sampling them jointly, all at once, is intractable.
- The state path alone can be hundreds of unknowns.

The easy split

- States given parameters: a Kalman problem.
- Parameters given states: a standard regression.
- Alternate the two, and the tangle dissolves.

The Two-Block Gibbs Sweep

States, then parameters, then repeat

block 1 (states): $\beta_{1:T} \sim p(\beta_{1:T} \mid \theta, y)$ (FFBS)
block 2 (params): $\theta \sim p(\theta \mid \beta_{1:T}, y)$

Block one

- Fix the parameters at their current draw.
- Draw the entire state path in one go (FFBS).
- This is the linear-Gaussian half.

Block two

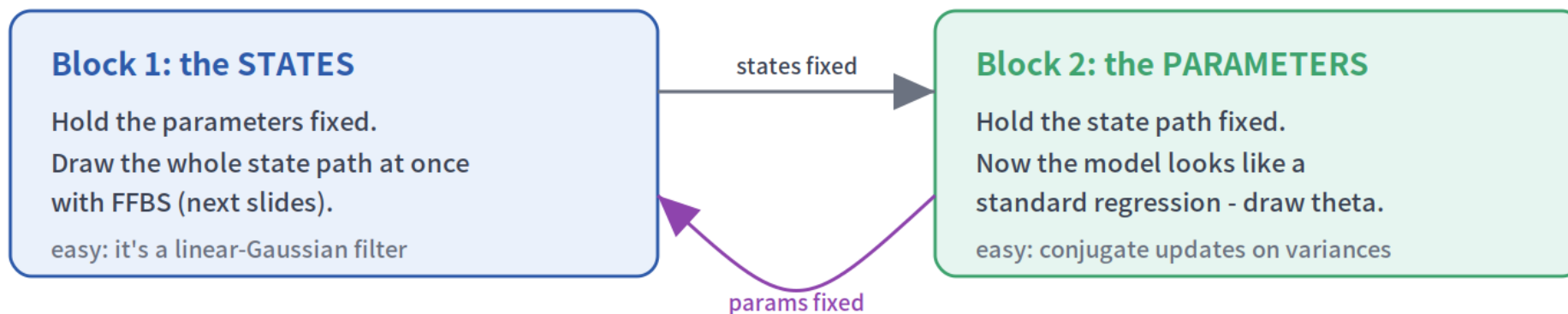
- Fix the state path just drawn.
- Draw the parameters from their conditional.
- For the variances, these are conjugate updates.

A Two-Block Gibbs Sweep

Hold the parameters and draw the whole state path with FFBS; then hold the states and draw the parameters as a regression. Each conditional is easy, and repeating the sweep explores the full joint posterior

A Two-Block Gibbs Sweep

Alternate: draw all the states given the parameters, then the parameters given the states.



Why split it in two

Jointly, states and parameters are a hard tangle. But EACH given the other is easy: states given parameters is a Kalman problem; parameters given states is a simple regression. Alternating explores the joint posterior.

Repeat the sweep thousands of times - the draws are the joint posterior.

Draw the States Given the Parameters

A clean linear-Gaussian problem

What makes it easy

- With parameters fixed, the model is linear-Gaussian.
- That is exactly the Kalman setting from Week 9.
- The whole state path can be drawn at once with FFBS.

What we get

- One complete, consistent draw of the hidden path.
- Not just a mean — an actual sampled trajectory.
- Fed forward to the parameter step.

Draw the Parameters Given the States

Now it's just a regression

What makes it easy

- With the state path fixed, the states are 'known'.
- The model collapses to a standard regression.
- The noise variances get simple conjugate updates.

What we get

- A fresh draw of the parameters.
- Consistent with the current state path.
- Fed back into the next state-sampling step.

Forward-Filter, Backward-Sample

How to draw a whole state path at once

Block one needs a full path, not just filtered means. FFBS does it in two passes: run the Kalman filter forward to gather information, then sweep backward, drawing each state given the one after it — yielding one coherent path per sweep.

Why not just filter

- The filter gives a mean and variance at each time.
- But we need a joint DRAW of the whole path.
- Independent per-time draws would be inconsistent.

What FFBS does

- Forward pass: the familiar Kalman filter.
- Backward pass: sample the path end to start.
- The result is one internally consistent trajectory.

The FFBS Factorization

Draw the end, then work backward

$$p(\beta_{1:T} \mid y, \theta) = p(\beta_T \mid y) \prod_{t=1}^{T-1} p(\beta_t \mid \beta_{t+1}, y_{1:t})$$

Reading it

- Start by drawing the final state from the filter.
- Then draw each earlier state given the next one.
- Each conditional uses the forward-filtered quantities.

Why it works

- The chain structure makes each step Gaussian.
- Conditioning on the next state ties the path together.
- One backward pass yields one full sample.

The Forward Filter Pass

Gather information, left to right

What it does

- Run the Kalman filter from start to finish.
- At each time, form the filtered mean and variance.
- Store them — the backward pass will need them.

What it is

- Exactly the filter from Week 9, unchanged.
- It summarizes all data up to each time.
- It is the 'learn' half of FFBS.

The Backward Sampling Pass

Draw a path, right to left

What it does

- Draw the final state from its filtered distribution.
- Step backward, drawing each state given the next.
- Use the stored forward quantities at each step.

What it gives

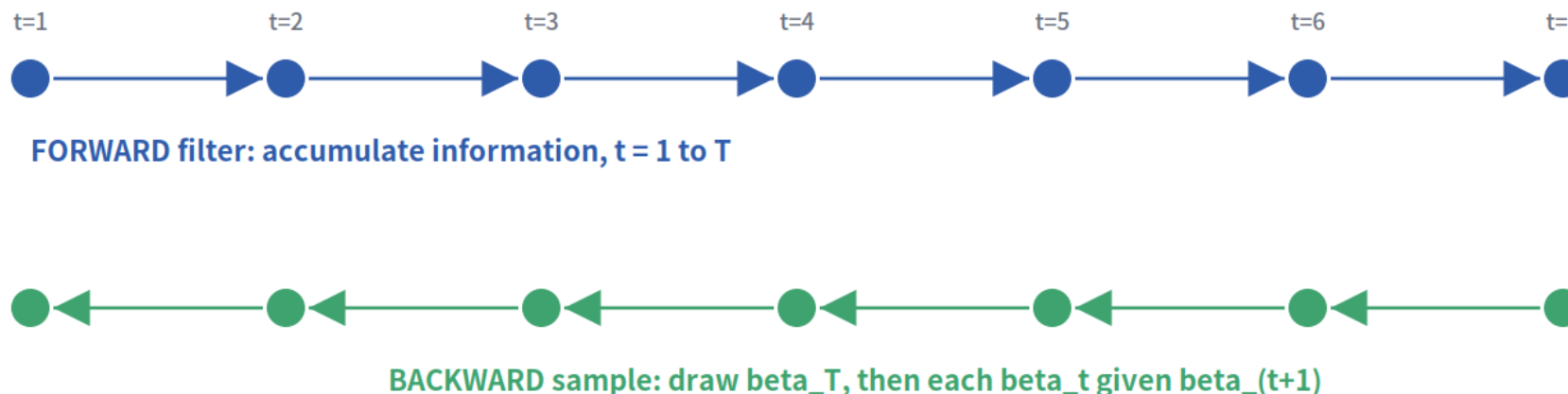
- One complete, coherent state path.
- A genuine sample, not a smoothed average.
- It is the 'draw' half of FFBS.

FFBS: Forward to Learn, Backward to Draw

The forward pass is the Kalman filter, accumulating information from the first observation to the last. The backward pass draws the final state, then each earlier one given its successor — one full path per sweep.

FFBS: Forward to Learn, Backward to Draw

One pass filters left-to-right; a second pass samples a coherent path right-to-left.



The forward pass is the familiar Kalman filter. The backward pass starts at the end and draws each state given the one after it - producing ONE complete, internally consistent path in a single sweep.

Each Sweep Draws One Path

Many sweeps make the fan

One sweep

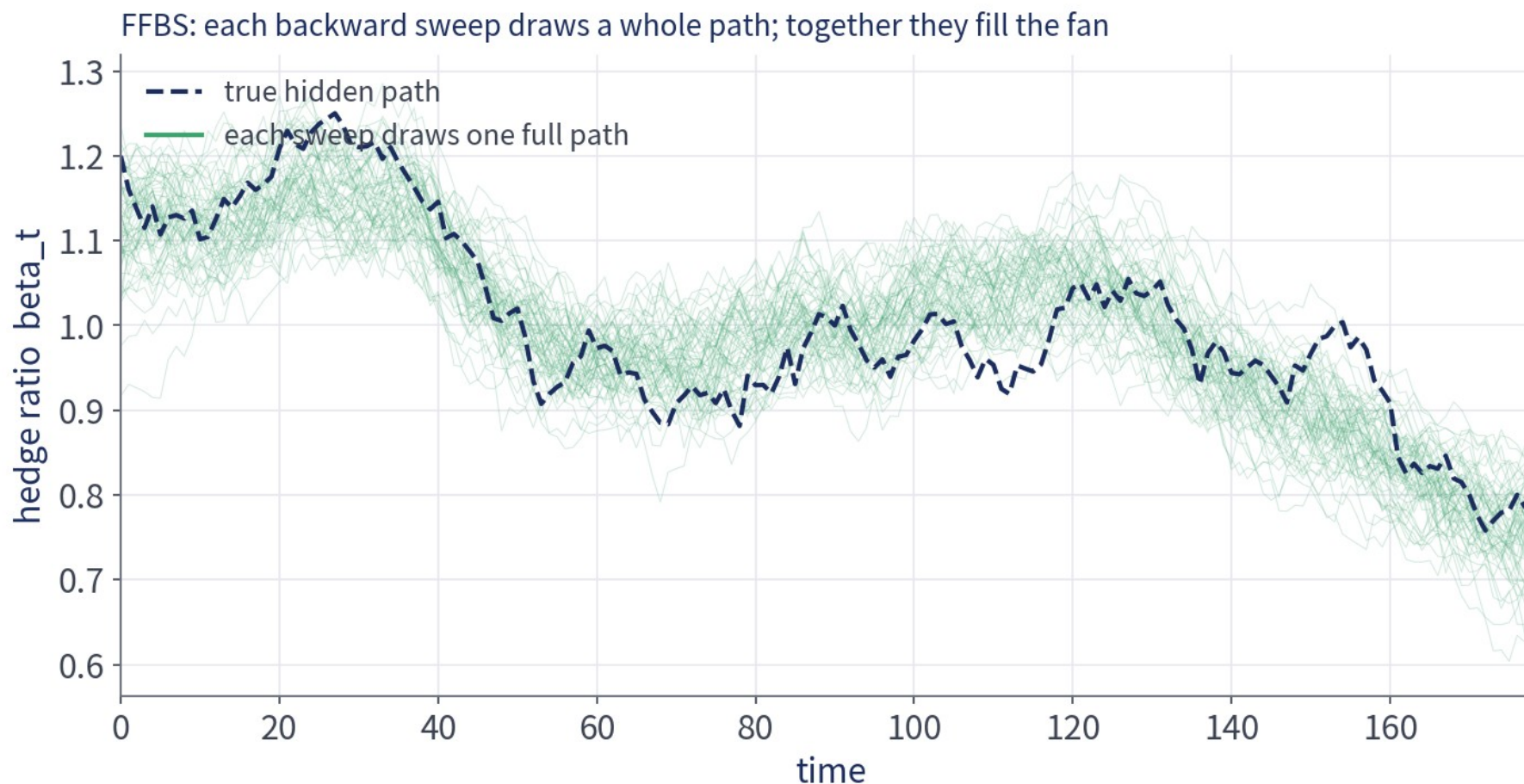
- A single FFBS pass yields one whole path.
- It is a plausible history of the hidden state.
- Different sweeps give different paths.

Many sweeps

- Collect hundreds of sampled paths.
- Together they trace out the posterior fan.
- Their spread at each time is the credible band.

FFBS Builds the Fan

Each backward sweep draws one complete hedge-ratio path. Overlaid, hundreds of these draws fill out the posterior fan around the true hidden path — the raw material for a credible band.



The Bayesian Time-Varying Hedge Ratio

Pairs trading Stage III, now with credible bands

Recall the pairs-trading hedge ratio from Stage III — the coefficient linking the two assets, which drifts over time. Modeled as a Bayesian state-space, it becomes a posterior fan: a median hedge ratio with an honest band at every date.

The Stage III object

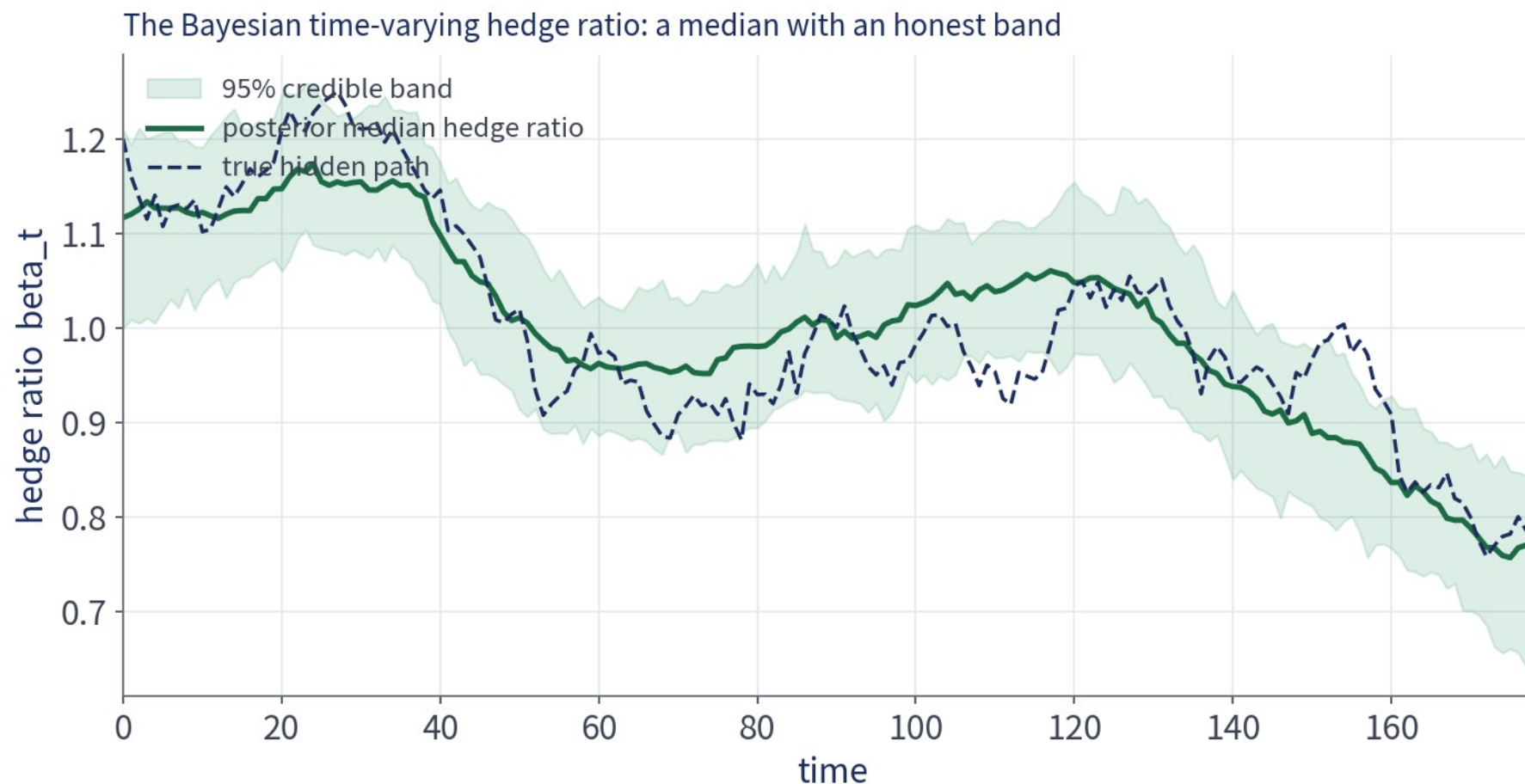
- The hedge ratio relating the two legs of the pair.
- It is not constant — it drifts as markets change.
- Earlier we tracked it with a single filtered line.

The Bayesian upgrade

- Now a posterior median plus a credible band.
- Wide where the data is uninformative.
- Honest about how sure the ratio really is.

The Hedge Ratio With a Credible Band

The posterior median hedge ratio, wrapped in a 95% credible band from the FFBS draws. The band widens when the data says less — exactly the honesty a single filtered line could never provide.



Trading Stage III, Now Honest

The band changes the signal

With the old line

- You used one hedge ratio, taken as exact.
- The spread it implied looked precise.
- You couldn't tell a firm signal from a shaky one.

With the band

- A tight band means a trustworthy hedge ratio.
- A wide band warns the relationship is uncertain.
- You can stand down when the pair is ill-defined.

TVP-SV: Adding Stochastic Volatility

When the noise level is itself a hidden, moving process

So far the noise level was constant. But financial volatility clusters and drifts. TVP-SV lets the variance itself be a hidden, evolving state — a time-varying coefficient on top, a breathing volatility underneath, both inferred at once.

The limitation so far

- We assumed a fixed observation-noise level.
- But market volatility is anything but fixed.
- It clusters — calm spells and turbulent spells.

The TVP-SV answer

- Add a second hidden state: the log-volatility.
- Let it drift like the coefficient does.
- Infer both moving processes jointly.

Why Constant Variance Isn't Enough

Volatility is a moving target

What constant variance misses

- It averages calm and crisis into one level.
- It understates risk in turbulent periods.
- And overstates it in quiet ones.

What a moving variance captures

- Bands that widen in turbulent stretches.
- Tighter inference when markets are calm.
- Risk estimates that track the real environment.

The TVP-SV Model

Two hidden processes, one system

$$\begin{aligned} y_t &= \beta_t x_t + e^{h_t/2} \varepsilon_t \\ \beta_t &= \beta_{t-1} + \eta_t, \quad h_t = h_{t-1} + \nu_t \end{aligned}$$

The two states

- The coefficient drifts as a random walk.
- The log-volatility drifts as its own random walk.
- The observation noise scales with that volatility.

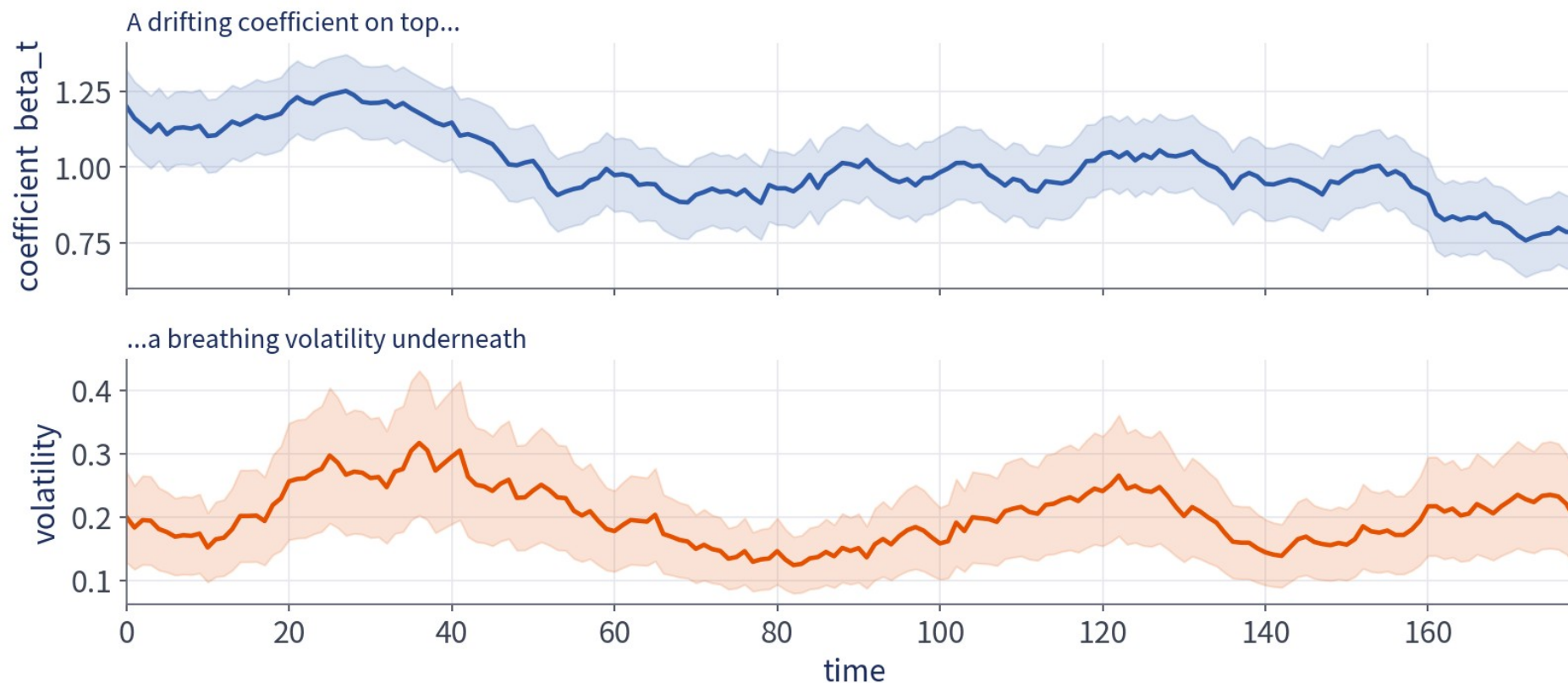
Inferring it

- The same Gibbs strategy, with an extra block.
- A dedicated sampler handles the volatility state.
- Out comes a joint posterior over both processes.

TVP-SV in One Picture

A drifting coefficient on top and a breathing volatility underneath, each with its own credible band. Both are hidden states, inferred together — the model adapts its own sense of risk over time.

TVP-SV: both the coefficient AND the noise level are hidden, moving processes



Bayesian Time-Varying Hedge Ratio

Direct Claude Code to fit a dynamic linear model in PyMC

GOAL Fit a drifting hedge ratio as a Bayesian state-space model and produce a credible band.

[INPUT] Prompt · Claude Code

Write and run `bayes_hedge.py` that:

1. simulates a pair whose hedge ratio drifts as a random walk (a local-level DLM);
2. specifies a Bayesian state-space model in PyMC with priors on the state and observation noise;
3. samples the joint posterior over the whole hedge-ratio path and the parameters;
4. plots the posterior median hedge ratio with a 95% credible band against the true path.

Show the band widens where the data is less informative and covers the truth.

How to work with it

- The hedge ratio is the hidden state.
- Priors go on the two noise levels.
- The output is a fan, not a line.
- Wide band = shaky hedge ratio.

Deliverable: `bayes_hedge.py` + a posterior-median hedge ratio with a credible band vs the truth.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What are the two blocks in the Gibbs sweep?

**States given params,
params given states**

Q2 What does one FFBS sweep produce?

**One full sampled
state path**

Q3 What extra hidden process does TVP-SV add?

**A drifting log-
volatility**

All three come from the two-block sweep, FFBS, and the stochastic-volatility state.

Three Pitfalls to Avoid

Where the sampling goes wrong

Common mistakes

- "I can draw each state independently from the filter."
- "FFBS gives a smoothed line."
- "Constant variance is fine for financial data."

The reality

- Independent per-time draws are inconsistent — use FFBS.
- FFBS draws a full sampled PATH, not a mean.
- Volatility drifts; TVP-SV models it as a hidden state.

Three Things to Carry Forward

Part 2 in one screen

1

Alternate states and parameters

A two-block Gibbs sweep draws the whole state path given the parameters, then the parameters given the states. Each conditional is easy; alternating explores the joint posterior.

2

FFBS draws a whole path at once

Forward-filter to gather information, then sample backward from the last state to the first. Each sweep yields one coherent path; many sweeps make the fan and its credible band.

3

The hedge ratio becomes a fan, and volatility can breathe

The Stage III hedge ratio gains an honest credible band. TVP-SV adds a drifting volatility, so the model tracks changing risk as a hidden state of its own.

Next — Part 3: the Bayesian HMM, soft regime probabilities, structural breaks, and pairs Stage VI.